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Inventor(s)

WEYER; Axel

Casting-rolling installation, and method for producing a steel strip

Abstract

A casting-rolling installation and a corresponding method serve for producing a steel strip that runs through the casting-rolling installation in a conveying direction. The casting-rolling installation includes a two-roller casting device with two casting rollers that are each arranged mounted rotatably about axes aligned axially parallel to one another and at the same height. A cast steel strip is produced by the two-roller casting device. A rolling mill stand for rolling the cast steel strip is mounted downstream of the casting device in the direction of conveyance of the steel strip. The rolling mill stand is arranged relative to the casting device in such a manner that a roll nip inlet of the rolling mill—as seen in the vertical direction—is located with respect to the position of the axes of the casting rollers at the same height or above same.

Inventors:	WEYER; Axel (Wuppertal, DE)
Applicant:	SMS group GmbH (Mönchengladbach, DE)
Family ID:	86328659
Assignee:	SMS group GmbH (Mönchengladbach, DE)
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a national stage application, filed under 35 U.S.C. § 371, of International Patent Application PCT/EP2023/060116, filed on Apr. 19, 2023, which claims the benefit of German Patent Application DE 10 2022 204 069.4, filed on Apr. 27, 2022.

TECHNICAL FIELD

[0002] The disclosure relates to a casting-rolling installation for producing a steel strip, and to a corresponding method for producing a steel strip.

BACKGROUND

[0003] In the prior art, a technology (device and method) for producing cast strip from a metal melt is known, for example from WO 2014/049150 A1. Thereby, a metal melt passes through a casting nip bounded by two counter-rotating casting rolls and is formed into the cast strip. When casting metal melts on devices that are also referred to as “twin-roll casting machines,” two axially parallel and internally cooled casting rolls in each case rotate in opposite directions to one another, delimiting the longitudinal sides of a casting nip between them. The narrow sides of the casting nip are usually sealed by plates made of a refractory material. In each case, enough liquid melt is cast into the casting nip such that a so-called “melt pool” forms above the casting nip and is maintained until the casting process is completed. The melt from the melt pool reaches the casting rollers and solidifies to form a shell in each case, which is then conveyed into the casting nip by the respective casting roll. The shells are then pressed against one another in the casting nip, such that the cast strip is formed from them and the melt trapped between them. The cast strip that exits continuously from the casting nip in this manner is drawn off below the casting rollers and fed to the roll nip of a rolling mill for further processing at a lower height level.

[0004] Further conventional casting-rolling installations with associated methods for producing steel strips are known, for example, in EP 2162251 B1 or CN 112522575 A.

[0005] A conventional casting-rolling installation under the prior art is shown in principle in simplified form in the side view of FIG. 8, which is a typical production installation for steel strip. The height difference between the casting rollers of the twin-roller and the mill floor of the rolling mill is, for example, 5000 mm.

[0006] Such a conventional installation in accordance with FIG. 8 has the disadvantage that, with respect to the casting device provided here, the position of the axes of the casting rollers—as seen in the vertical direction—is always arranged considerably higher than a roll nip inlet of a rolling mill that is directly connected to the casting device. As a result, the previous solutions have the disadvantage that the costs for the hall construction, for foundations and the associated steel structures, e.g. for the elevation of the continuous casting installation, represent a comparatively large factor for an investment in a new installation or an extension of existing steelworks installations due to the different height structure of the casting installation and rolling mill. The associated crane systems, lighting, ventilation, entrances and access routes for operating equipment are also adversely affected by the height structure and arrangement of installations. For the operating personnel (maintenance, servicing and operation of the installation), the use of the necessary staircases, the use of decentralized control points and in the restricted access routes for the delivery of operating equipment due to the given height differences of the installations are an ongoing hindrance to the upcoming work tasks and lead to disadvantageously high operating costs.

SUMMARY

[0007] Accordingly, the disclosure is based on the object of optimizing a casting-rolling installation and an associated method for producing a steel strip and thereby reducing both the necessary construction/production capital expenditures (CAPEX) of the overall installation and its ongoing operational expenditures (OPEX) for the production of strips from steel.

[0008] The above-mentioned object is achieved by a casting-rolling installation as disclosed herein and by a method for producing a steel strip as disclosed herein.

[0009] A casting-rolling installation serves for producing a steel strip that runs through the casting-rolling installation in a conveying direction. Such a casting-rolling installation comprises a casting device, which is designed in the form of a two-roller casting device and accordingly has two casting rollers that are in each case arranged mounted rotatably about axes aligned parallel to one another and at the same height, wherein a cast steel strip is produced by means of the two-roller casting device, and a rolling mill stand for rolling the cast steel strip is mounted downstream of the casting device in the conveying direction.

[0010] In accordance with one embodiment, with such a casting-rolling installation it is provided that the rolling mill stand is arranged relative to the casting device in such a manner that a roll nip inlet of the rolling mill—as seen in the vertical direction—is located with respect to the position of the axes of the casting rollers at the same height or above same.

[0011] In accordance with an alternative embodiment, with such a casting-rolling installation it is provided that the rolling mill stand is arranged relative to the casting device in such a manner that a roll nip inlet of the rolling mill stand—as seen in the vertical direction—is located with respect to the position of the axes of the casting rollers below it, and the distance of the roll nip inlet from the position of the axes—as seen in the vertical direction—assumes a distance s , which fulfills the following condition: $s \leq (2 \times D)$, where D denotes a diameter of a casting roller.

[0012] In the same manner, the disclosure also provides a method for producing a steel strip, which can be carried out with a casting-rolling installation as explained above. In any case, with this method, it is provided that a steel strip is cast by a casting device in the form of a two-roller casting device, and is subsequently moved in a conveying direction to a rolling mill stand mounted downstream for rolling the cast steel strip. Here, a loop, which is formed by the cast steel strip between the casting device and the rolling mill stand, is set to a predetermined contour/position, preferably in a controlled manner.

[0013] The disclosure is based on the substantial insight that, with a casting-rolling installation, the characteristic positioning/arrangement of a rolling mill stand provided directly adjacent to the casting device relative to the position of the axes of the casting rollers thereof, the resulting overall height of such an installation is advantageously reduced.

[0014] The method for producing a steel strip is characterized in that a predetermined contour/position is always achieved/set for a loop formed by the cast steel strip between the casting device and the rolling mill stand, in particular by a suitable setting/control of a rotational speed of the casting rollers and/or a rolling speed of the rolling mill stand, in order to achieve both a high operational reliability of the installation and a high production quality for the produced steel strip.

[0015] In an advantageous further development, the casting rollers are designed in such a manner that the width of the steel strip cast with them is $\leq 2,200$ mm. In other words, with casting rollers designed in this manner, a steel strip with a width of up to 2,200 mm can be cast.

[0016] In an advantageous further development, the casting rollers are designed in such a manner that their diameter is $\leq 1,500$ mm. In this connection, it should be noted that the two casting rollers, which are part of a two-roller casting device, expediently have the same diameter.

[0017] In an advantageous further development, a loop detection device is provided, by means of which a position of the cast steel strip and a loop formed thereby, which is located between the casting device and the rolling mill stand, can be determined. Such a loop detection device comprises at least one non-contact distance sensor, which can be designed in the form of a laser

and/or a video camera. In any case, by means of such a loop detection device, it is possible during the operation of the casting-rolling installation/when carrying out the method, to appropriately detect/recognize the contour of a loop formed by the cast steel strip between the casting device and the rolling mill stand. As already explained elsewhere, a predetermined contour with respect to this loop can then be realized by a suitable setting/control of a rotational speed of the casting rollers and/or a rolling speed of the rolling mill stand.

[0018] In an advantageous further development, a control unit is provided which is connected to the loop detection device by signal technology. Here, the control unit is configured by program technology in such a manner that a rotational speed of the casting rollers and/or a rolling speed of the rolling mill stand can be controlled in such a manner that the loop formed by the cast steel strip assumes a predetermined contour.

[0019] Taking into account the above-mentioned control unit, when the method for producing a steel strip is carried out, it is achieved that, as a function of the contour/position of the loop formed by the cast steel strip detected by the loop detection device, a rotational speed of the casting rollers and/or a rolling speed of the rolling mill stand are set/changed in a controlled manner, so that the resulting loop formed by the cast steel strip assumes a predetermined contour.

[0020] In an advantageous further development of the method, it can be provided that a rotational speed of the casting rollers about their axes is selected in such a manner that the resulting casting speed for the cast steel strip assumes a value between 20-100 m/min.

[0021] Further advantages of the present design consist of the following aspects/features: [0022] Cost reduction of the construction and production costs of the installations, in particular by reducing the required hall height. [0023] Installation of the casting device and a downstream rolling mill on a common level, for example in the form of a mill floor or a corresponding working platform, as a result of which the construction costs are reduced and subsequent use is also associated with lower operating costs. [0024] Cost reduction in operating costs. [0025] If the production of liquid steel for supplying the casting device in the form of the twin-roller casting device, e.g. via a plurality of smaller induction furnaces (preferably with a capacity of 5-10 tons), the upstream region for the steel production process can also be lowered to a lower height level for the entire hall construction (steelworks, twin roller and rolling mill).

[0026] Exemplary embodiments of the invention are described in detail below with reference to a schematically simplified drawing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 shows a casting-rolling installation for producing a steel strip in accordance with a first embodiment.

[0028] FIG. 2 shows further details of the casting-rolling installation of FIG. 1, namely with regard to an arrangement of the associated casting device and a rolling mill stand mounted downstream relative to one another in the conveying direction.

[0029] FIG. 3 shows the casting device of the installation in FIG. 1 for the production of a strip-shaped material from steel melt.

[0030] FIG. 4 shows the arrangement of a casting device and a rolling mill stand mounted downstream relative to one another in the conveying direction for a casting-rolling installation in accordance with a second embodiment.

[0031] FIG. 5 shows the arrangement of a casting device and a rolling mill stand mounted downstream relative to one another in the conveying direction for a casting-rolling installation in accordance with a third embodiment.

[0032] FIG. 6 shows a loop detection device, with which a casting-rolling installation can be

equipped.

[0033] FIG. 7 shows further details with respect to a possible supply of liquid steel to the casting device of a casting-rolling installation.

[0034] FIG. 8 shows a conventional installation.

DETAILED DESCRIPTION

[0035] With reference to FIGS. 1-7, preferred embodiments of a device **10** and a corresponding method for producing a steel strip are explained below. Identical features in the drawing are in each case marked with the same reference signs. At this point, it is separately pointed out that the drawing is only simplified and not to scale.

[0036] FIG. 1 schematically shows a side view of a casting-rolling installation **10**, with which a steel strip **1** can be produced.

[0037] The casting-rolling installation **10** comprises a casting device **12**, which is designed in the form of a two-roller casting device and accordingly has two casting rollers **13**. The casting rolls **13** are in each case arranged mounted rotatably on axes (X1, X2) that are aligned axially parallel to one another and at the same height (see FIG. 3).

[0038] A casting nip G is formed between the casting rollers **13**, which casting nip can be seen in the end-face view of the casting rollers **13** in FIG. 3. For providing a liquid steel melt, a pan **2** and an intermediate container **3** are provided for the installation **10** of FIG. 1, wherein a liquid steel melt is introduced into the casting nip G from above via an outlet of the intermediate container **3**. The liquid steel melt can then exit vertically downward through this casting nip G, in order to solidify here into a cast steel strip **1**.

[0039] Furthermore, the casting-rolling installation **10** comprises a rolling mill stand **14** for rolling the cast steel strip **1** mounted downstream of the casting device **12** in the conveying direction F.

[0040] After exiting from the casting nip G, the cast steel strip **1** is moved in a conveying direction through the casting-rolling installation **10**. This conveying direction is symbolized by an arrow and denoted with “F” in FIG. 1.

[0041] A transport of the cast steel strip **1** through the rolling mill stand **14** is ensured by drivers T, which are arranged in each case upstream and downstream of the rolling mill stand **14**, as seen in the conveying direction F.

[0042] Downstream of the rolling mill stand **14**, the casting-rolling installation **10** is further equipped with a cooling section **24**, at least one shear **26** and a final reel **28**, on which the steel strip **1** produced can be wound in a known manner.

[0043] Between the casting device **12** and the rolling mill stand **14**, which is arranged downstream thereto in the conveying direction F, the casting-rolling installation **10** has at least one turning roller **16**. This turning roller **16** is arranged in such a manner that the cast steel strip **1** is deflected in the direction of the rolling mill stand **14**.

[0044] The above-mentioned turning roller **16** is significant in that the cast steel strip **1**, after it has exited vertically downwards from the casting nip G, initially takes the form of a loop **19**, which is formed by the cast steel strip **1** between the casting device **12** and the rolling mill stand **14**. This loop **19** is then deflected horizontally in a controlled manner in the direction of the rolling mill **14** by means of the turning roller **16**. It is important here that the turning roller **16** is located at approximately the same height as a roll nip inlet of the rolling mill stand **14**, such that the turning roller **16** allows the steel strip **1** to run horizontally into the roll nip inlet of the rolling mill stand **14**.

[0045] With respect to the casting-rolling installation **10**, it should be emphasized that its substantial components are formed by the casting device **12** and the downstream rolling mill stand **14**. The arrangement of the casting device **12** and the rolling mill stand **14** relative to one another—as seen in the vertical direction—is of substantial importance. This is explained in detail below with reference to FIGS. 2, 4 and 5.

[0046] FIG. 2 illustrates an arrangement of the casting device **12** and the rolling mill stand **14**

relative to one another, in accordance with a first embodiment. The casting device **12** and the rolling mill stand **14** are in each case attached/set up on a common mill floor H (or a comparable area). Here, the axes X1, X2 of the casting rollers **13** are spaced from the mill floor H by a distance a, wherein a roll nip inlet **15** of the rolling mill stand **14** is spaced from the mill floor by a distance b.

[0047] By having the two distances a and b in each case assume the same value with the first embodiment in accordance with FIG. 2, i.e. the axes X1 and X2 of the two casting rolls **13** on one hand, and the roll nip inlet **15** of the rolling mill stand **14** on the other hand, are in each case spaced equally from/above the mill floor H, it is ensured that the roll nip inlet **15** of the rolling mill **14**—as seen in the vertical direction—is at the same height with respect to the position of the axes X1, X2 of the casting rollers **13**. Here, it is also important that the turning roller **16**, as already explained, is located at approximately the same height as the roll nip inlet **15** of the rolling mill **14**. With respect to axes X1, X2 of casting rollers **13** and the roll nip of the rolling mill stand **14**, an equal level is thus achieved—as seen in the vertical direction.

[0048] FIG. 4 illustrates an arrangement of the casting device **12** and the rolling mill stand **14** relative to one another, in accordance with a second embodiment. Here, the distance a, by which the axes X1, X2 of the casting rollers **13** are spaced from the mill floor, H is greater than the distance b, by which the roll nip inlet **15** of the rolling mill stand **14** is spaced from the mill floor H. As a result, the roll nip inlet **15** of the rolling mill stand **14**—as seen in the vertical direction—is located above it with respect to the position of the axes X1, X2 of the casting rollers **13**. This in turn results in the fact that the distance of the roll nip inlet **15** from the position of the axes X1, X2—as seen in the vertical direction—assumes a distance s that fulfills the following condition:

$s \leq (2 \times D)$, [0049] with D=diameter of a casting roller **13**.

[0050] The diameter of a casting roller is illustrated in FIG. 3.

[0051] With the representation of the second embodiment in accordance with FIG. 4, the distance s is approximately half the value of the diameter of a casting roller, i.e. $s = 0.5 \times D$. It is important for this second embodiment that the arrangement of the casting device **12** and the rolling mill stand **14** relative to one another is selected in such a manner that the distance s, as explained, does not become greater than twice the value of the diameter of a casting roller **13**.

[0052] FIG. 5 illustrates an arrangement of the casting device **12** and the rolling mill stand **14** relative to one another, in accordance with a third embodiment. Here, the distance a, by which the axes X1, X2 of the casting rollers **13** are spaced from the mill floor, H is smaller than the distance b, by which the roll nip inlet **15** of the rolling mill stand **14** is spaced from the mill floor H. As a result, the roll nip inlet **15** of the rolling mill stand **14**—as seen in the vertical direction—is located above it with respect to the position of the axes X1, X2 of the casting rollers **13**.

[0053] At this point, it is pointed out separately that, with the second and third embodiments in FIGS. 4 and 5, both the complete loop **19** of the cast steel belt **1** and the turning roller **16** are not shown in order to simplify the representation.

[0054] With respect to the above-mentioned first, second and third embodiments of the casting-rolling installation **10**, it is pointed out separately at this point that the characteristic arrangement of the casting device **12** and the rolling mill stand **14** relative to one another achieves a reduction in the height difference (see FIG. 2) or even a complete elimination (see FIG. 2) of this height difference between the axes of rotation of the casting rollers **13** and the roll nip inlet of the rolling mill stand **14** mounted downstream. This advantageously achieves a lower overall height for the casting-rolling installation **10**/a smaller vertical distance between the casting device **12** and the rolling mill stand **14** and a common mill floor H. As already explained at the beginning, this can reduce the required hall height and further construction costs for the casting-rolling installation **10**.

[0055] FIGS. 6 and 7 show and explain further options that can be provided in the same manner in the first, second and third embodiments of the casting-rolling installation **10**.

[0056] In accordance with the representation in FIG. 6, a loop detection device **18** is preferably arranged adjacent to an underside of the casting device **12**, by means of which a position of the cast steel strip **1** and the loop **19** formed thereby, which is located between the casting device **12** and the rolling mill stand **14**, can be determined. Such a loop detection device **18** comprises at least one non-contact distance sensor **20**, which can be designed in the form of a laser and/or a video camera.

[0057] In conjunction with the loop detection device **18**, the casting-rolling installation **10** in accordance with FIG. 6 is also equipped with a control unit R, which is connected to the loop detection device **18** by signal technology. Accordingly, by means of the control unit R and as a function of the contour/position of the loop **19** formed by the cast steel strip **1** detected by the loop detection device **18**, it is possible to set/change the rotational speed of the casting rollers **13** and/or the rolling speed of the rolling mill stand **14** in a controlled manner, such that the resulting loop **19** formed by the cast steel strip **1** assumes a predetermined contour.

[0058] In accordance with the representation of FIG. 7, instead of a pan **1**, it is possible that—as seen in the conveying direction F—the heating is effected upstream of the casting device **12** by means of at least one induction furnace **22**. In such an induction furnace **22**, the starting material **4**, for example scrap metal, is heated and melted. The steel melt **5** formed in this way is subsequently suitably introduced from above into the casting nip G of the casting device **12** formed between the two casting rollers **13**.

[0059] In the representation of FIG. 7, two such induction furnaces **22** are shown, with which a steel melt can be provided/produced for the casting device **12** as explained. As an alternative to the representation in FIG. 7, only one induction furnace **22** can be provided. Irrespective of the number of such induction furnaces **22**, it may be emphasized that a further advantageous reduction of the overall height of the casting-rolling installation **10**/of its height level above a mill floor H or the like is hereby achieved.

LIST OF REFERENCE SIGNS

[0060] **1** Steel strip [0061] **2** Pan [0062] **3** Intermediate container [0063] **4** Starting material [0064] **5** Steel melt [0065] **10** Casting-rolling installation [0066] **12** Casting device [0067] **13** Casting roller [0068] **14** Rolling mill stand [0069] **15** Roll nip inlet [0070] **16** Turning roller [0071] **18** Loop detection device [0072] **19** Loop (formed by a cast steel strip **1**) [0073] **20** Non-contact distance sensor [0074] **21** Signal section [0075] **22** Induction furnace [0076] **24** Cooling section [0077] **26** Shear [0078] **28** Reel [0079] D Diameter (of a casting roller **13**) [0080] F Conveying direction [0081] G Casting nip [0082] H Mill floor [0083] R Control unit [0084] S Distance or spacing—seen in the vertical direction—between roll nip inlet **15** and axis (X1, X2) of a casting roller **13** [0085] T Driver [0086] X1, X2 Axis (of a casting roller **13**)

Claims

1.-14. (canceled)

15. A casting-rolling installation (**10**), comprising a two-roller casting device (**12**) having two casting rollers (**13**) that are mounted rotatably about axes (X1, X2) which are aligned parallel to one another and arranged at a same height, the two-roller casting device (**12**) being configured to produce a cast steel strip (**1**); and a rolling mill stand (**14**) for rolling the cast steel strip (**1**) mounted downstream of the two-roller casting device (**12**) in a conveying direction (F), the rolling mill stand (**14**) being arranged relative to the two-roller casting device (**12**) such that a roll nip inlet (**15**) of the rolling mill stand (**14**) is located vertically at or above the same height of the axes (X1, X2) of the casting rollers (**13**).

16. The casting-rolling installation (**10**) according to claim 15, further comprising a turning roller (**16**) arranged between the two-roller casting device (**12**) and the rolling mill stand (**14**), the turning roller (**16**) being configured deflect the cast steel strip (**1**) towards the rolling mill stand (**14**).

17. The casting-rolling installation (**10**) according to claim 15, wherein the casting rollers (**13**) are

designed such that a width of the cast steel strip (1) is $\leq 2,200$ mm.

18. The casting-rolling installation (10) according to claim 15, wherein a diameter of the casting rollers (13) is $\leq 1,500$ mm.

19. The casting-rolling installation (10) according to claim 15, further comprising a loop detection device (18) for determining a position of a loop (19) formed by the cast steel strip (1), the loop (19) being located between the two-roller casting device (12) and the rolling mill stand (14).

20. The casting-rolling installation (10) according to claim 19, wherein the loop detection device (18) comprises a non-contact distance sensor (20).

21. The casting-rolling installation (10) according to claim 20, wherein the non-contact distance sensor (20) includes a laser and/or a video camera.

22. The casting-rolling installation (10) according to claim 19, further comprising a control unit (R) that is operatively connected to the loop detection device (18), wherein the control unit (R) is configured to control a rotational speed of the casting rollers (13) and/or a rolling speed of the rolling mill stand (14) such that the loop (19) assumes a predetermined contour.

23. The casting-rolling installation (10) according to claim 15, further comprising at least one induction furnace (22) arranged above the two-roller casting device (12) for feeding liquid steel melt into a casting nip (G) formed between the casting rollers (13).

24. A method for producing a steel strip, comprising: casting the steel strip (1) by a two-roller casting device (12) having two casting rollers (13); subsequently moving the steel strip (1) in a conveying direction (F) to a rolling mill stand (14) mounted downstream for rolling the steel strip (1); controlling a rotational speed of the casting rollers (13) and/or a rolling speed of the rolling mill stand (14) such that a loop (19) formed by the steel strip (1) between the casting device (12) and the rolling mill stand (14) assumes a predetermined contour; and controlling the rotational speed of the casting rollers (13) so that the steel strip (1) assumed a casting speed between 20 m/min and 100 m/min.

25. The method according to claim 24, further comprising detecting an actual contour of the loop (19) by a loop detection device (18).

26. The method according to claim 25, wherein controlling the rotational speed of the casting rollers (13) and/or the rolling speed of the rolling mill stand (14) is performed by a control unit (R) that is operatively connected to the loop detection device (18) as a function of the actual contour of the loop (19) and the predetermined contour.
